

JIRA AGILE LAB 2

Backlog Creation, Story Point Estimation & Sprint Execution

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Project	Digital Campus Library Reservation Gateway
Use Case	UC-04 – Pay Overdue Fine
Sprint	Sprint 1

Final Submission

1. Objective

The objective of this lab is to convert the functional requirement represented by UC-04 “Pay Overdue Fine” into an Agile backlog, create and estimate user stories, prioritize the backlog, plan Sprint 1, execute the sprint on a Jira board, and analyze sprint progress using the Burndown Chart.

2. Project and Use Case

Project: Digital Campus Library Reservation Gateway. The selected use case is UC-04 – Pay Overdue Fine. The primary actor is the Student Member and the supporting external actor is the Payment Gateway.

The successful flow is: the student views outstanding itemized fines, selects a payment method, the system connects to the Payment Gateway, the payment is authorized, the fine is marked paid, the balance becomes zero, and a receipt is generated and emailed. A declined payment leaves the fine outstanding and allows another payment attempt.

3. Epic and User Stories

Epic: EPIC-01 – Overdue Fine Payment & Settlement. The epic covers viewing the fine, choosing a payment method, completing payment, generating a receipt, and handling failed payments.

Jira ID	User Story	Priority	Points
DCLRG-2	View Outstanding Fine	High	3
DCLRG-3	Select Payment Method	High	2
DCLRG-4	Pay Overdue Fine	High	5
DCLRG-5	Generate Payment Receipt	Medium	3
DCLRG-6	Handle Declined Payment	High	5
DCLRG-7	Maintain Fine Balance on Failure	Medium	3

Total backlog estimate: 21 story points.

4. User Story Details

DCLRG-2 – View Outstanding Fine

As a student member, I want to view my outstanding fine balance and itemized fines per book, so that I know exactly what I owe before making a payment.

DCLRG-3 – Select Payment Method

As a student member, I want to choose a payment method such as card, UPI, or campus wallet, so that I can pay my overdue fine using an available option.

DCLRG-4 – Pay Overdue Fine

As a student member, I want to pay my overdue fine through the Payment Gateway, so that my outstanding balance is settled.

DCLRG-5 – Generate Payment Receipt

As a student member, I want to receive a payment receipt after successful payment, so that I have confirmation of the transaction.

DCLRG-6 – Handle Declined Payment

As a student member, I want the system to inform me when a payment is declined and allow another payment method, so that I can retry the payment.

DCLRG-7 – Maintain Fine Balance on Failure

As a student member, I want my fine balance to remain unchanged when payment fails, so that unsuccessful transactions do not incorrectly clear my debt.

5. Jira Epic and Backlog Evidence

The Jira epic contains the six child work items created from UC-04. The screenshot shows the user stories, priorities, story points, and their association with the epic.

The screenshot displays a Jira Epic page for 'Overdue Fine Payment & Settlement'. The page includes a description, attachments, and a list of child work items. The child work items are as follows:

Work	Prior...	Story ...	Assi...	Status
☒ DCLRG-2 View Outstanding Fine	^ H...	3	Uni	To Do
☒ DCLRG-3 Select Payment Method	^ H...	2	Uni	To Do
☒ DCLRG-4 Pay Overdue Fine	^ H...	5	Uni	To Do
☒ DCLRG-5 Generate Payment Receipt	= M...	3	Uni	To Do
☒ DCLRG-6 Handle Declined Payment	^ H...	5	Uni	To Do
☒ DCLRG-7 Maintain Fine Balance on Failure	= M...	3	Uni	To Do

Below the table, there is a section for 'Linked work items' with a link to 'Add linked work item'.

Figure 1. Jira Epic – Overdue Fine Payment & Settlement with six child user stories.

6. Sprint 1 Planning

Sprint 1 was planned around the successful overdue-fine payment flow. Four stories were selected for the sprint, giving a planned capacity of 13 story points: DCLRG-2 (3), DCLRG-3 (2), DCLRG-4 (5), and DCLRG-5 (3). The remaining failure-handling stories were retained in the product backlog for subsequent work.

7. Sprint 1 Execution

The Active Sprint board was used to move the selected stories through To Do, In Progress, and Done. At the end of the execution, all four planned stories were completed.

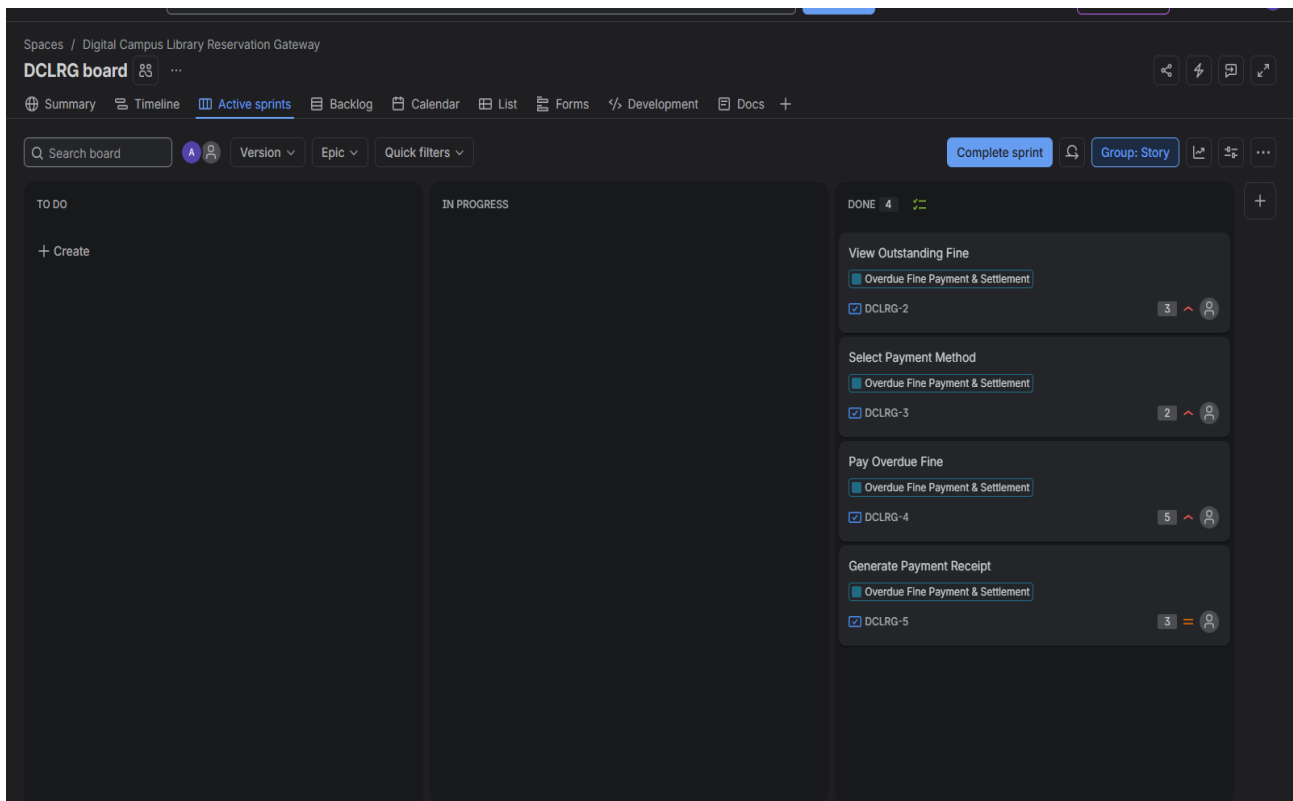


Figure 2. Jira Active Sprint – all four Sprint 1 stories completed.

8. Sprint 1 Burndown Analysis

The Burndown Chart records Sprint 1 progress using story points. The sprint started with 13 story points remaining. As each selected story was completed, the remaining work decreased from 13 to 10, 8, 3, and finally 0 story points.

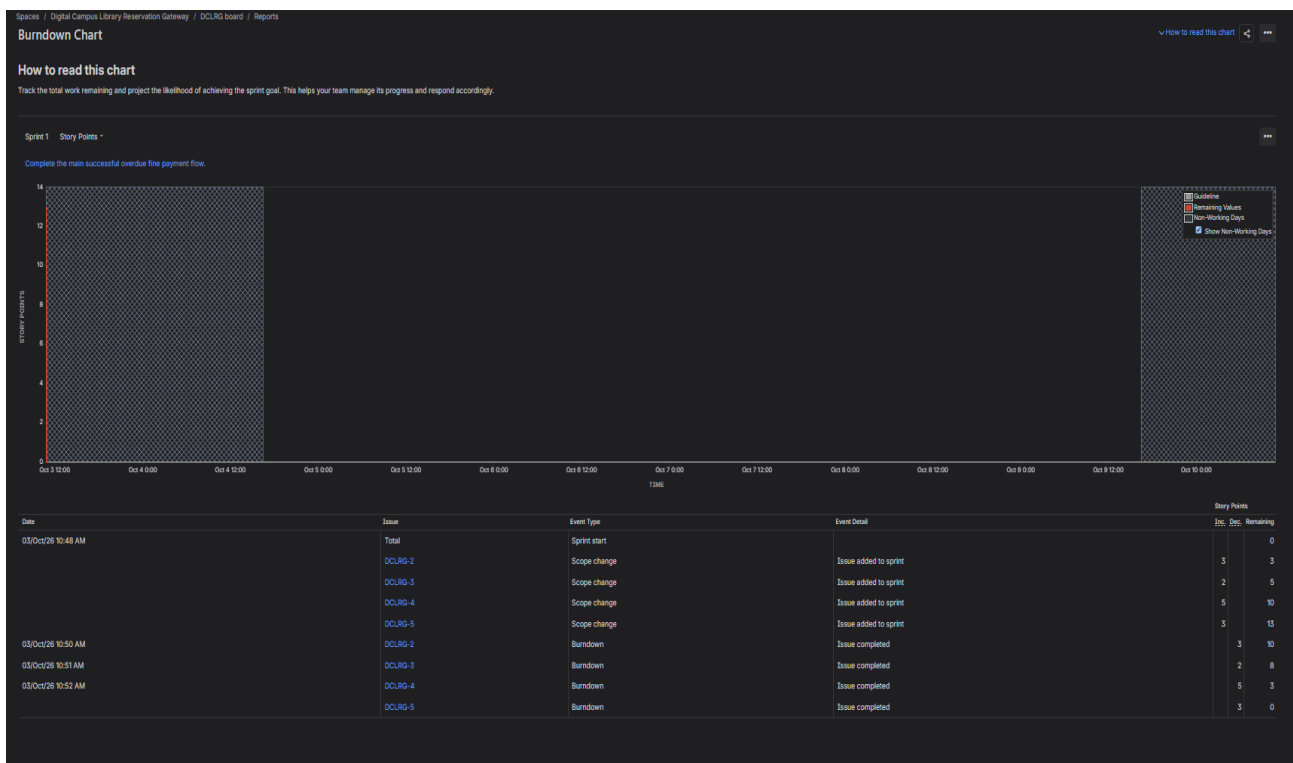


Figure 3. Sprint 1 Burndown Chart showing completion from 13 to 0 story points.

9. Reflection

1. Did the estimates reflect the actual effort?

The Fibonacci story-point estimates provided a relative indication of effort. The 5-point Pay Overdue Fine story involved more functionality because it required the payment flow and interaction with the external Payment Gateway, while smaller supporting stories were estimated at 2 or 3 points.

2. Was the backlog prioritized effectively?

Yes. The sprint focused first on the core successful payment flow: viewing the fine, selecting a payment method, completing payment, and generating a receipt. The remaining failure-handling stories were left for later work.

3. Was the sprint aligned with the requirements?

Yes. The four Sprint 1 stories directly represent the main successful scenario of UC-04 – Pay Overdue Fine and cover the flow from viewing the outstanding amount through successful payment and receipt generation.

4. What did the Burndown Chart show about capacity?

The Burndown Chart shows that Sprint 1 began with 13 story points and ended with 0 remaining points. All four planned stories were completed, so the planned sprint workload was fully burned down.

10. Conclusion

The Jira Agile workflow successfully converted UC-04 into an actionable backlog, estimated the work using story points, selected a focused Sprint 1 workload, executed the stories on the Scrum board, and analyzed progress through the Burndown Chart. The sprint finished with all 13 planned story points completed.